Preparing Data for Analysis

Turning Raw Data into Useful Information

Lecture 3 • Module 1: Data Literacy & Python

Press Space to start →

Module 1

Preparing Data for Analysis

3.1 Where We Are

The workflow so far

```
Find Data
↓
Inspect Data
↓
⭐ Prepare Data
↓
Visualize Data
```

Today's shift

We are moving from **understanding** the dataset to **improving** it. The goal is to shape raw stock data into something easier to analyze.

Big idea: good analysis usually starts with data that was prepared for a specific question.

3.2 Learning Objectives

By the end of this lecture, students can...

- Create new columns from existing stock data
- Explain what a derived feature is
- Select useful columns for analysis

And also...

- Build a smaller analysis dataset
- Save processed data to a new CSV file
- Explain why raw data is not always enough

3.3 Why Raw Data Isn't Enough

Dataset A: downloaded raw data

Column
Date
Open
High
Low
Close
Volume

Dataset B: prepared for a question

Column
Date
Close
Daily Return
7-Day Average

This version is smaller, more focused, and easier to use for charting or comparison.

3.4 A New Idea

Feature engineering

Data scientists often create **new information** from existing columns. These new columns help answer questions more directly.

Simple pipeline

Raw data

↓

Better data

↓

Better decisions

3.5 First New Feature

Notebook cell

```
df["Price Difference"] = df["Close"] - df["Open"]
```

What it tells us

- Was the stock higher at the end of the day?
- How far did price move between open and close?
- Which days were stronger or weaker?

Class question: what does a **positive** price difference tell us?

3.6 Another Useful Feature

Notebook cell

```
df["Daily Return"] = df["Close"].pct_change()
```

Why this feature matters

- Did the stock increase or decrease?
- By how much, in percentage terms?
- Can we compare different days fairly?

Daily return is often more useful than price alone because it measures **relative change**.

3.7 More Feature Ideas

Original Column	Possible New Feature	Why It Helps
Close	Daily Return	Measures percentage change
Open, Close	Price Difference	Shows intraday direction
Close	7-Day Average	Smooths short-term noise
Volume	Volume Change	Shows trading activity shifts

Good analysts do not just collect data. They create information that fits the question.

3.8 Choosing Useful Columns

Notebook cell

```
df[["Date", "Close", "Daily Return"]]
```

Why select fewer columns?

- the dataset becomes easier to read
- charts become simpler to build
- we focus on the information that answers the question

3.9 Different Questions Need Different Data

Question	Useful Columns
How is the stock trending over time?	Date, Close
When was trading activity unusually high?	Date, Volume
Which days had the biggest moves?	Date, Daily Return

3.10 Building an Analysis Dataset

1. Download CSV
2. Inspect the data
3. Create features
4. Select columns
5. Save the analysis dataset

3.11 Save Your Dataset

Notebook cell

```
df.to_csv("stock_analysis.csv", index=False)
```

Professional habit

- keep the raw data
- keep the processed data
- keep the final analysis dataset

Never overwrite the original file. If something goes wrong, you need the raw version again.

3.12 Class Activity

Student task

Choose your own company

Create **Price Difference**

Create **Daily Return**

Keep these columns: Date, Close, Daily Return, Volume

Save it as

```
company_analysis.csv
```

The goal is to create a smaller dataset that is ready for charts in the next lecture.

Summary

What we did

- Created new information from existing data
- Selected useful columns
- Built an analysis dataset
- Saved a processed CSV file

Big idea

We transformed **raw data** into **useful information**. That makes the next steps in analysis much easier.

Next Lecture

Visualizing Stock Data

Turning numbers into charts with **Matplotlib**.